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Deeplearning

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problem-1

$x = [5.4, 3.0, 1.5, 0.2]$

$w_1 = 1, w_2 = 3, w_3 = 5, w_4 = 7$

$b = 1.5$

$z = x_1 w_1 + x_2 w_2 + x_3 w_5 + x_4 w_7 + b$

Solution :-

$z = (5.4 \times 1) + (3.0 \times 3) + (1.5 \times 5) + (0.2 \times 7) + 1.5$

$= 5.4 + 9 + 7.5 + 1.4 + 1.5$

$\boxed{z = 24.8}$

Apply Sigmoid

$\sigma(z) = \frac{1}{1+e^{-z}}$

$= \frac{1}{1+e^{-24.8}}$

$\sigma(z) \approx 1$

$h = 1$

Sigmoid :-

$\sigma(1.3) = \frac{1}{1+e^{-1.3}}$

$\Rightarrow 0.786$

versicolor :-

$w = 1.5, b = 0.2$

$z_{versico} = (1.5) + 0.2$

$\boxed{1.7}$

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$\sigma(1.7) \approx \boxed{0.846}$

Virginica

$w = 1.8, b = 0.1$

$z_{virginica} = 1(1.8) + 0.1$

1.9

$\sigma(1.9) \approx 0.870$

Output layer:-

$w = 1.2 \quad b = 0.1$

$z_{setosa} = h(1.2) + 0.1$

$= 1(1.2) + 0.1$

$\boxed{z_{setosa} = 1.3}$

Problem-2

$X = [6.1, 2.8, 4.0, 1.3]$

$w_1 = 1, \; w_2 = 2, \; w_3 = 4, \; w_4 = 6$

$b = -0.5$

Formula:-

$z = x_1 w_1 + x_2 w_3 + x_3 w_5 + x_4 w_7 + b$

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Solution:-

Hidden layer

$Z = (6.1 \times 1) + (2.8 \times 2) + (4.0 \times 4) + (1.3 \times 6) - 0.5$

$= 6.1 + 5.6 + 16 + 7.8 - 0.5$

$= 35.5 - 0.5$

$Z = 35.0$

Sigmoid

$\sigma(z) = \frac{1}{1+e^{-z}}$

$= \frac{1}{1+e^{-35}}$

$\sigma(35) \approx 1.$

$h = 1$

Setosa

$Z_{setosa} = 1(1.1) + 0.2$

$= 1.3$

$\sigma(1.3) = 0.786$

Vessicolor

$Z_{vessicolor} = 1(1.4) + 0.1$

$Z_{vessicolor} = 1.5$

$\sigma(1.5) = 0.818$

Virginica

$Z_{virginica} = 1(1.7) + 0.3$

$Z_{virginica} = 2.0$

$\sigma(2.0) = 0.887$

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Problem - 3

$X = [6.7, 3.1, 5.6, 2.4]$

$w_1 = 1, w_2 = 2, w_3 = 3, w_4 = 4$

$b = -2$

formula

$$z = x_1 w_1 + x_2 w_3 + x_3 w_5 + x_4 w_7 - b$$

Solution:-

$$z = (6.7 \times 1) + (3.1 \times 2) + (5.6) \times 3) + (2.4 \times 4) - 2$$
$$= 6.7 + 6.2 + 16.8 + 9.6 - 2$$
$$= 39.3 - 2$$
$$z = 37.3$$

Sigmoid:-

$$\sigma(37.3) = \frac{1}{1 + e^{-z}}$$

$$= \frac{1}{1 + e^{-37.3}} = 1$$

$h = 1$

Output layer:-

Setosa

Versicolor

$z_{versicolor} = 1(1.6) + 0.2$

$= 1.8$

$\sigma(1.8) = 0.858$

Virginica

$z_{virginica} = 1(1.9) + 0.15$

$z_{virginica} = 2.05$

$\sigma(2.05) = \boxed{0.886}$

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$z_{setosa} = 1(1.3) + 0.1$

$z_{setosa} = 1.4$

$\sigma(1.4) = 0.802_4$